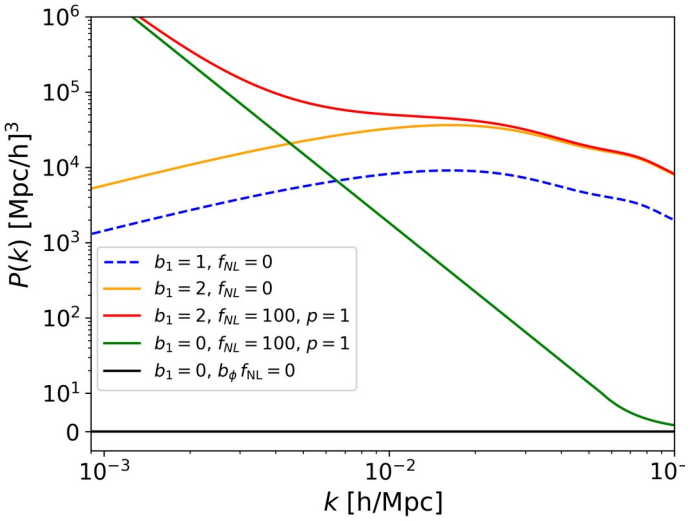
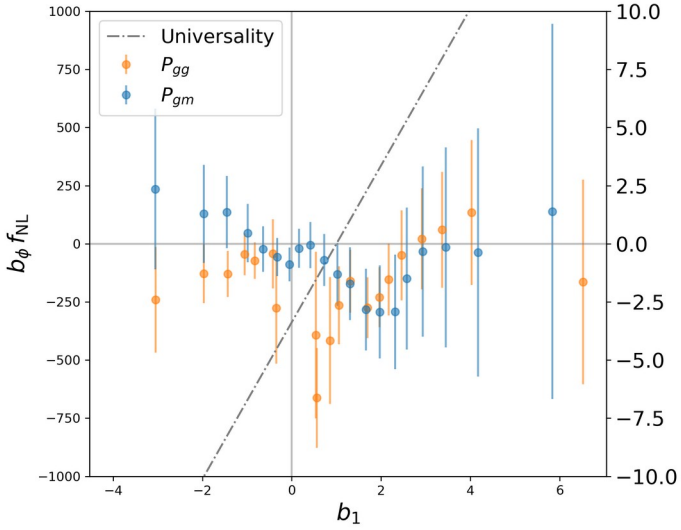
The lack of PNG response of zero-bias tracers

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Objective: test the viability of local halo overdensity classification to constraint PNG using zero-bias tracers, based on the proposal of Castorina *et al.* (2018), using the PNG-UNITsim suite (Adame *et al*, 2024).





$$P_{gg} = b^2 P_{mm}$$

$$b(k) = b_1 + b_\phi f_{\rm NL} \underbrace{\mathcal{M}^{-1}(k)}_{\propto k^{-2}}$$

$$\sigma(f_{\rm NL})^{-2} \propto \frac{{b_1}^2 {b_\phi}^2}{(b^2 P_{mm}(k) + 1/n)}$$

Results: local halo overdensity classification is not optimal for the detection of PNGs, as the results obtained are compatible with $b_\phi \simeq 0$.